Sapphire Pulse RX 7900 GRE Gaming OC 16GB Graphics Card

Golden Rabbit Edition

An exclusive to the Mainland China market until now, the AMD Radeon RX 7900 GRE (Golden Rabbit Edition) is a slightly scaled down version of the 7900 made to celebrate the year of the Rabbit – 2023. Well, it's no longer the year of the rabbit and the card is no longer an exclusive as AMD has now launched it globally. There is no vanilla Radeon RX 7900 so the GRE isn't a new term that AMD is going to be using across the stack like the XT or XTX suffixes that it has. For all intents and purposes, think of this as the RX 7900 that should have launched along with the other top-end cards.

The AMD Radeon RX 7900 GRE serves as the middle ground between the RX 7800 XT and the RX 7900 XT. And you might be wondering why AMD is releasing a card that has already been released? Well, since NVIDIA has been rather busy padding its roster with cheaper cards, it was only natural for AMD to also put a few cards out. Which also begs the question, is this too little and too late? Let's find out.

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SPECIFICATIONS

The AMD Radeon RX 7900 GRE is based on the Navi 31 GPU which is the same that powers the Radeon RX 7900 XTX and the XT. The only difference is the number of active Dual Compute Units within the Shader Engines. Navi 31 has a total of six Shader Engines which can host up to 6144 Shaders and the GPU that sits on the 7900 GRE is a slightly cut down version of that. We don't know if it has five or six active Shader Engines since the 40 Dual Compute Units can fit in either configuration. The 7900 GRE has a total of 80 Compute Units or 5120 Shaders which churn out 45.98 TFLOPs of performance at peak clock speeds. And since each Compute Unit hosts one Ray Tracing core, we have a total of 80 Ray Tracing cores. The other components such as the Render Output Units (ROP) and Texture Units are proportional i.e. 192 and 320, respectively.

Coming to the memory configuration, the AMD Radeon RX 7900 GRE has a similar configuration as the RX 7800 XT. So it has 16 GB of GDDR6 memory chips connected to a 256-bit wide memory bus. The memory clock is at 2250 MHz which is a bit lower than the RX 7800 XT. So peak memory data rate is 18 GBps as opposed to the 19.5 GBps bandwidth available on the 7800 XT. And the clock speeds are also a little lower at 2245 MHz boost clock as opposed to the 2438 MHz that the 7800 XT is at. More shader cores running at higher clock speeds can often consume a lot of power so this is absolutely normal. Until you see the clock speeds on the 7900 XT or XTX which are clocked at 2394 MHz and 2499 MHz despite having way higher shader cores.

Essentially, the AMD Radeon RX 7900 GRE has fewer cores which ideally could have been clocked higher but hasn't been. And it has slower memory than all the cards in its vicinity which is also a step back. All of this leads to about 40 Watts of power savings, if at all that was the end goal. It seems like there is a lot of potential in this card that cannot be leveraged or will not be leveraged. All of that is totally fine if this is priced right and performs competitively. So, let's see how it fares in that department.

PERFORMANCE

Compared to the RX 6800 XT, the 7900 GRE is a minor improvement over the 6800 XT. We did not have a 7800 XT to compare the card against, so we can't tell you how much of a performance difference there exists between them. We have the scores for the 7900 XTX, which is about 43 per cent ahead in synthetic benchmarks. And compared to the 6800 XT, it is about 22 per cent ahead. In gaming, the 7900 GRE is about 4.5-5 per cent ahead of the 6800 XT. And